

Bill Of Material(For Solar System)

Health Facility Name & Address

Project Details

Name: karnataka PHC

Project Number: PHC 02

Address: No. 2

Date: 2026-04-01

State: karnataka

Block: Belagavi

Sl. No.	Product	Make	Capacity	Quantity
1	Solar Module			
2	Solar Battery			
3	Module Mounting structure (MMS)	Technosphere	5 kW	10
4	Solar Charge Controller (CCU)	Technosphere	5 kW	10
5	DC -DC Converter	Technosphere	5 kW	10
6	Copper Cable (Module – CCU) -PV1-F (Solar cables)	Technosphere	5 kW	10
7	Copper Cable (Battery - Battery) - (DC Cables)	Technosphere	5 kW	10
8	Copper Cable (Battery - CCU) - (DC Cables)	Technosphere	5 kW	10
9	Battery trolley box with wheels – Hard Plastic	Technosphere	5 kW	10
10	MC4 connector with inline fuse	Technosphere	5 kW	10
11	MC4 connectors	Technosphere	5 kW	10
12	MC4 connectors Y Branch	Technosphere	5 kW	10
13	Double Pole MCB (load Side) with Conduit box	Technosphere	5 kW	Sample Load Side Mcb Conduit Box Qty
14	Single Line Diagram (SLD) for the system	Technosphere	5 kW	10
15	Do's and Don'ts Practices Poster (Solar Panels, Battery and CCU)	Technosphere	5 kW	10
16	Signboard for Danger (Electric Shock), No Fire and PASS	Technosphere	5 kW	10
17	Fire Extinguisher	Technosphere	5 kW	10

Sl. No.	Product	Make	Capacity	Quantity
18	Metallic enclosure with isolator's having minimum gap of 1 inch(PV and Battery)	Technosphere	5 kW	10
19	Consumables	Technosphere	5 kW	10

Bill of material (For Luminaries & Fans)

Sl. No.	Product	Make	Capacity	Quantity	Remarks
1	LED Bulb	Technosphere	5 kW	10	Sample remarks
2	LED Bulb	Technosphere	5 kW	10	Sample remarks
3	LED Tube light	Technosphere	5 kW	10	Sample remarks
4	LED Tube light	Technosphere	5 kW	10	Sample remarks
5	Wall Mounted Fan	Technosphere	5 kW	10	Sample remarks
6	Mobile Charging with USB port	Technosphere	5 kW	10	Sample remarks

Bill of material (For Load Wiring)

Sl. No.	Items	Description & Make	Capacity	Quantity
1	Switch (Modular)	Sample Switch Modular Description : Technosphere	5 kW	10
2	Socket (Modular)	Sample Socket Modular Description : Technosphere	5 kW	10
3	USB Port for mobile charging	Sample Usb Port For Mobile Charging Description : Technosphere	5 kW	10
4	Cables - For Load Connection (Red)	Sample Cables For Load Connection Red	5 kW	10

Sl. No.	Items	Description & Make	Capacity	Quantity
		Description : Technosphere		
5	Cables - For Load Connection (Black)	Sample Cables For Load Connection Black Description : Technosphere	5 kW	10
6	Power Cable - From CCU to Room (Red)	:		
7	Power Cable - From CCU to Room (Black)	Sample Power Cable From Ccu To Room Black Description : Technosphere	5 kW	10
8	Ceiling Rose	Sample Ceiling Rose Description : Technosphere	5 kW	10
9	Angle holder	Sample Angle Holder Description : Technosphere	5 kW	10
10	1 modular Switch Box with plate	Sample 1 Modular Switch Box With Plate Description : Technosphere	5 kW	10
11	2 modular Switch Box with plate	Sample 2 Modular Switch Box With Plate Description : Technosphere	5 kW	10
12	3 modular Switch Box with plate	Sample 3 Modular Switch Box With Plate Description : Technosphere	5 kW	10
13	UPVC Conduit Pipe (White)	Sample Upvc Conduit Pipe White Description : Technosphere	5 kW	10
14	UPVC - Coupler (White)	Sample Upvc Coupler White Description : Technosphere	5 kW	10
15	UPVC Conduit Tee Joint	Sample Upvc Conduit Tee Joint Description : Technosphere	5 kW	10

Sl. No.	Items	Description & Make	Capacity	Quantity
16	UPVC - Short & Long Elbow (White)	Sample Upvc Short And Long Elbow White Description : Technosphere	5 kW	10
17	2way Junction Box	Sample 2Way Junction Box Description : Technosphere	5 kW	10
18	3way Junction Box	Sample 3Way Junction Box Description : Technosphere	5 kW	10
19	Square Box	Sample Square Box Description : Technosphere	5 kW	10
20	Plastic wall lug	10 : Technosphere	5 kW	Sample Plastic Wall Lug Description
21	Plastic wall lug	Sample Plastic Wall Lug 2 Description : Technosphere	5 kW	10
22	Screw	Sample Screw Description : Technosphere	5 kW	10
23	Screw	Sample Screw 2 Description : Technosphere	5 kW	10
24	Electrical Insulating Tape	Sample Electrical Insulating Tape Description : Technosphere	5 kW	10
25	Pipe Saddle Clamps	Sample Pipe Saddle Clamps Description : Technosphere	5 kW	10
26	Saddle Nail	Sample Saddle Nail Description : Technosphere	5 kW	10
27	Cable Tie	Sample Cable Tie Description : Technosphere	5 kW	10
28	Cable Lugs	Sample Cable Lugs Description :	5 kW	10

Sl. No.	Items	Description & Make	Capacity	Quantity
		Technosphere		
29	Flexible Pipe	Sample Flexible Pipe Description : Technosphere	5 kW	10
30	Labelling Tags (Load identification tags)	Sample Labelling Tags Load Identification Tags Description : Technosphere	5 kW	10
31	Labelling Tags (Cable identification tags)	Sample Labelling Tags Cable Identification Tags Description : Technosphere	5 kW	10

SYSTEM FUNCTIONALITY PARAMETERS

Date of recording	2026-04-01	Time of recording	NA
Weather Condition at the time of recording	Clear Sky		

Array Details

Array Size:(kWp)	4	No. of modules:	4	No. of strings:	4	No. of modules per string:	4
Each module capacity(Wp):	5 kW						

Array Parameters (CCU Input Side)

Test Condition	Voltage	Measured Value	Current	Measured Value
Measurement with Panel MCB OFF	V	String 1: 12.5	NA	NA
Measurement with Panel MCB OFF	V	String 2: 12.5	NA	NA
Measurement with Panel MCB OFF	V	Array: 12.5	NA	NA

Test Condition	Voltage	Measured Value	Current	Measured Value
Measurement with Panel MCB ON(Wait for 20 seconds)	Vmp(in Volts)	String 1:	A	String 1: 12.5
Measurement with Panel MCB ON(Wait for 20 seconds)	Vmp(in Volts)	String 2:	A	String 2: 12.5
Measurement with Panel MCB ON(Wait for 20 seconds)	Vmp(in Volts)	Array:	A	Array: 12.5
Measurement with Panel MCB ON(Wait for 20 seconds)	Pmp(in Volts)	String 1:	NA	NA
Measurement with Panel MCB ON(Wait for 20 seconds)	Pmp(in Volts)	String 2:	NA	NA
Measurement with Panel MCB ON(Wait for 20 seconds)	Pmp(in Volts)	Array:	NA	NA

Battery Bank parameters With Grid OFF & Load ON

Particulars	Measured Value	Unit
Battery Bank Voltage	230	V
Battery Bank Current	10	A

CCU parameters (Load ON)

Particulars	Measured Value	Unit
Load voltage/CGU output voltage	230	V
CCU output current at full load:(All solar loads turned on for 5 minutes)	10	A

CCU display & setting

Particulars	Set Value	Unit
Load-disconnect Voltage	12.0	V
Load-Reconnect Voltage	12.0	V
Battery Settings	Tubular	

Orientation Checklist

Checklist	Status	Remarks
Is the orientation on solar system's basic operational instructions given to the health center POC?	YES	Sample remarks
Is the orientation on solar system's basic routine maintenances given to the health center POC?	YES	Sample remarks
Are the stickers containing POCs pasted in the control room?	YES	Sample remarks
Are the Do's and Don'ts posters with helpline numbers pasted in the control room?	YES	Sample remarks
Additional remarks:	Sample remarks	